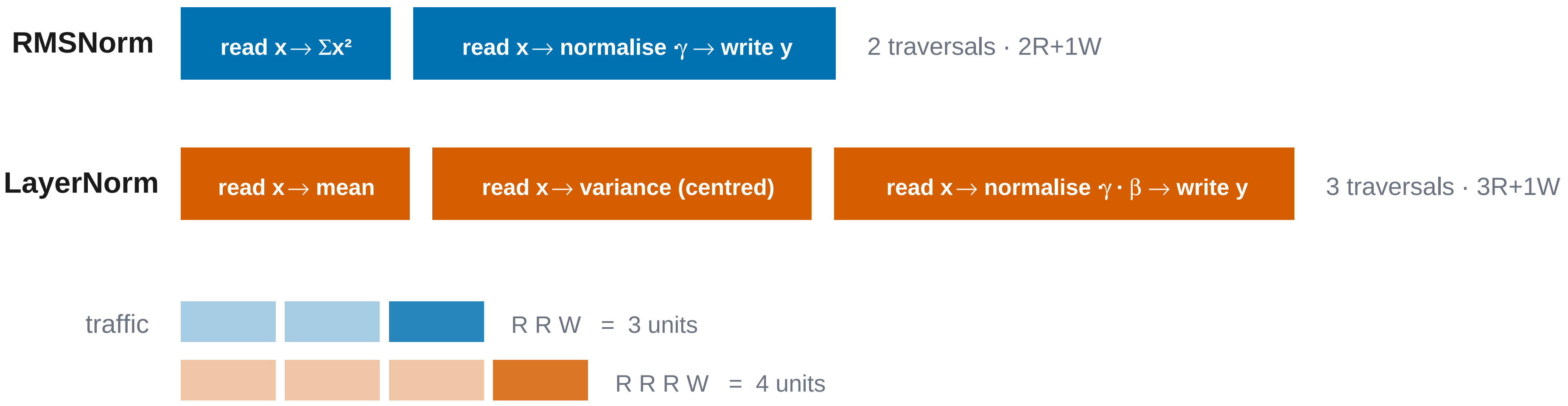
Why RMSNorm is faster: one fewer reduction stage

“Two-pass” and “single-pass” name the reduction stages; what costs bandwidth is the number of input traversals.



4 ÷ 3 = 1.33× more traffic per element for LayerNorm — the structural source of the gap.